DOCUMENT 01 · TIER 0–1

Foundation Fixes — the Production Floor

Defects in the current RENT CARS spec + the reliability patterns that turn it from a demo into a product. Build P0 first.

A1 · Tenant isolation — two layers, not one P0

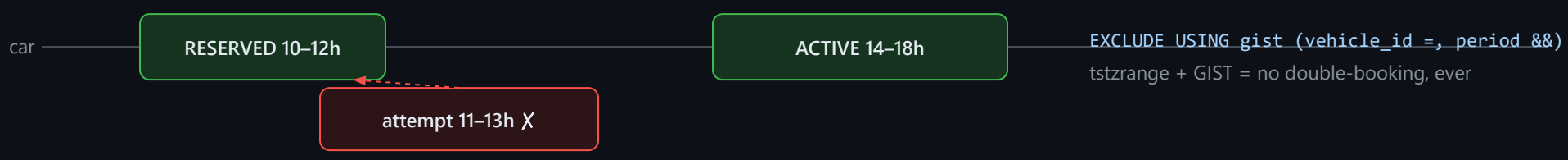


⚠ Pooler trap (CVE-2024-10976)

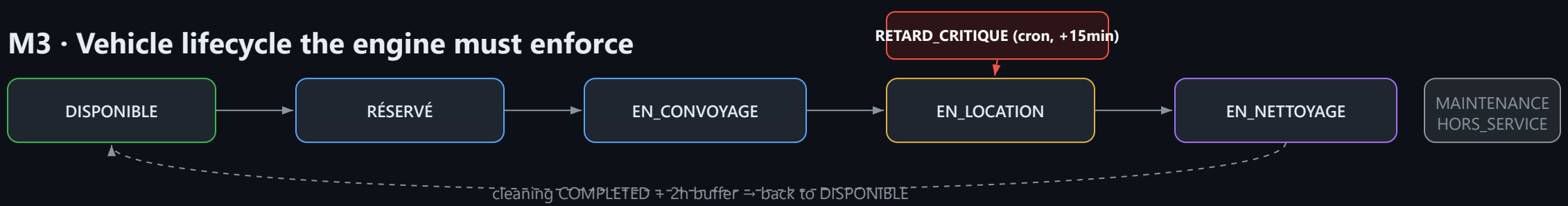
PgBouncer transaction mode leaks session state between tenants — bind it transaction-scoped: `set_config('app.current_tenant_id', tid, true)`

A2 · Overbooking made impossible at the database P0

One vehicle's timeline — Postgres rejects any overlapping booking, even under a race between counter agent & online user.



M3 · Vehicle lifecycle the engine must enforce



B1 · Transactional Outbox — no lost WhatsApp / debt events P1



Build order

P0 · ship first

- RLS + tx-scoped tenant ctx
- GIST anti-overbooking
- Webhook HMAC verify
- Durable idempotency (money)
- Refresh-token rotation

P1 · reliability

- Outbox + generalized DLQ
- Booking saga + compensation
- Double-entry ledger
- invoices / payments tables
- Missing tables (audit, tokens…)

P2 · scale & sight

- TimescaleDB for telemetry
- Tracing + SLOs + metrics
- Per-tenant rate limiting
- CI tenant-isolation test

P3 · only when forced

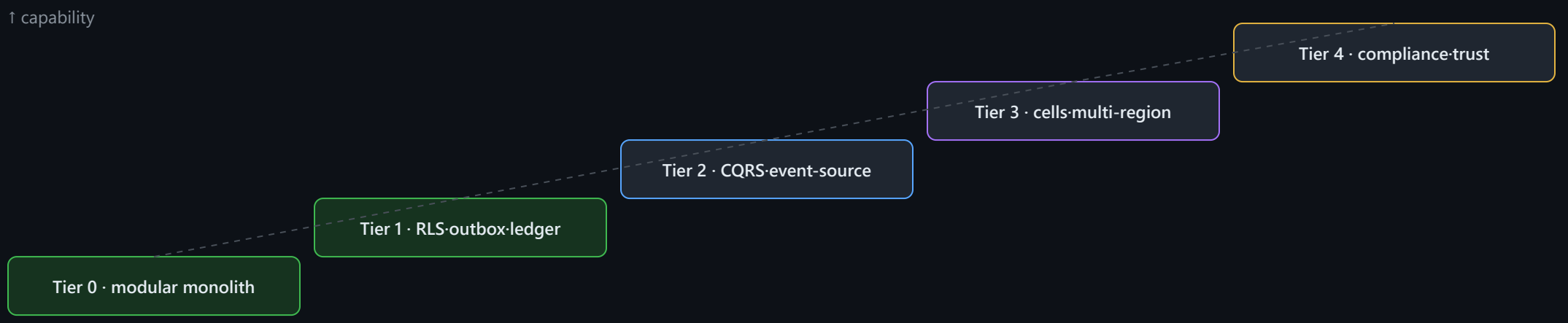
- Read replicas
- Partition big tables
- Isolate enterprise tenants
- Shard by tenant_id (last)

DOCUMENT 02 · TIER 2–4

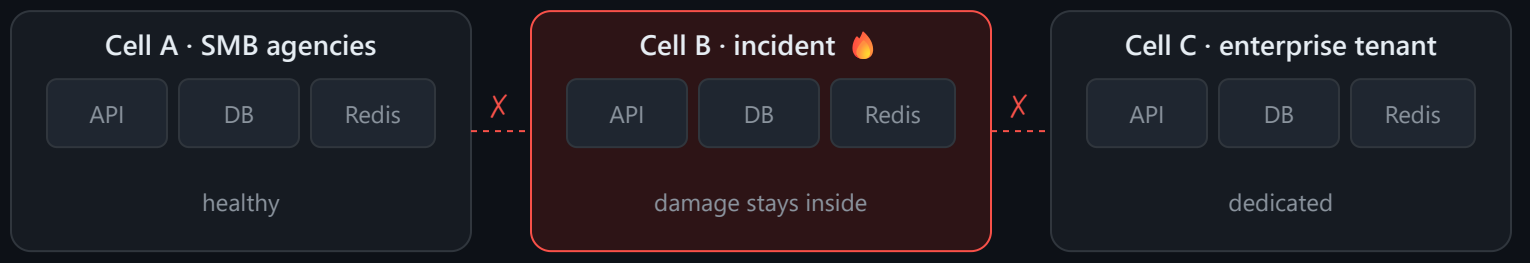
Scale, Resilience & the Enterprise Ceiling

Everything above the floor. None of it built early — the rule is climb only when a trigger metric forces it.

The maturity ladder — know the whole mountain, climb slowly



Cell-based architecture — contain the blast radius (Tier 3)

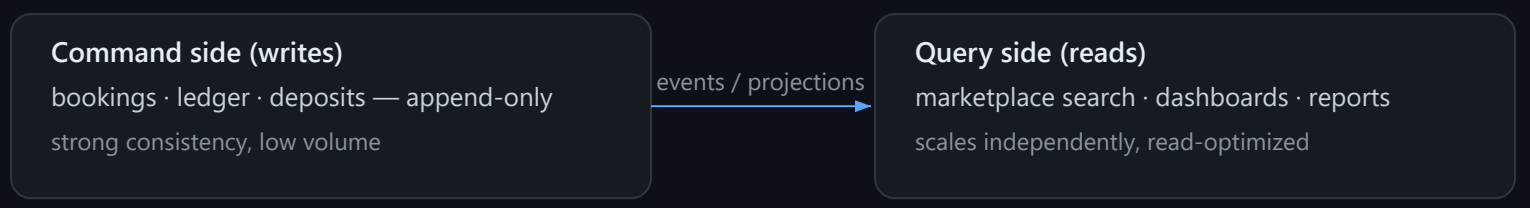


⚠ The 2026 way teams break cells

Shared deployment progression — same code to every cell at once collapses the isolation.

→ deploy cell-by-cell (canary across cells).

CQRS — split the read marketplace from the write money-path (don't do this in year 1)



Event sourcing — surgical use

bookings + ledger as a replayable log = court-grade dispute evidence & time-travel.

Trust & compliance layer — mandatory, not optional (Tier 4)



PCI-DSS → SAQ-A
PAN never touches your servers; tokenize

Crypto-shredding
"right to erasure" vs append-only ledger

Law 09-08 / CNDP
consent · GPS = personal data · GDPR-re

Hash-chained audit
tamper-evident = the real "sealed/immutable"

Climb a tier only when its trigger fires

Upgrade	Build it when…
Read replicas	/search p99 degrades or read CPU saturates primary
Partitioning	a table crosses ~50–100M rows / vacuum pain
TimescaleDB split	telemetry rows dominate DB size
CQRS	reads & writes provably contend, can't tune apart
Cells	a noisy tenant causes a cross-tenant incident
Multi-region	a contract requires data residency / failover SLA

DOCUMENT 03 · THE GAME

Disruption Thesis — New Rules for Car Rental

Incumbents sell a system of record. RENTY sells a system of trust, intelligence & network — attacking the unit economics.

The five wounds agencies actually bleed from

1 · Industrial fraud

AI fake IDs, stolen cards, synthetic identities.

Independents have zero structural defense.

2 · Insurance

The #1 survival bottleneck for

independents — decides if a fleet is even viable.

3 · Idle fleet

~75% of economics is fleet cycling now.

No yield tooling = lost spread on parked cars.

4 · Deposit / damage

Trust collapse BOTH ways: frozen €500–1000

+ false AI damage bills. No two-sided standard.

5 · Fragmented market

Morocco: 48% online by 2029, no trust layer.

Relationship dies the moment the contract is signed.

Why incumbents can't fix it — islands vs a network

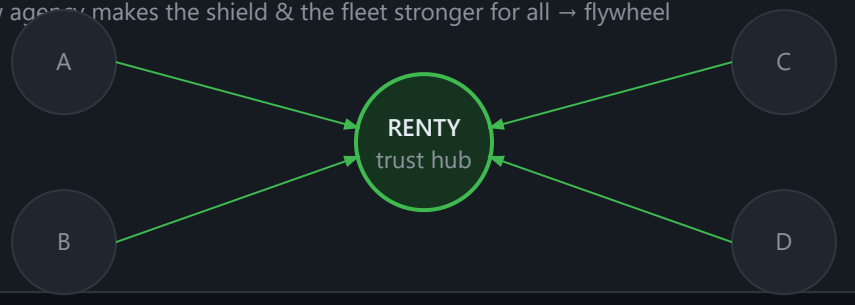
HQ Rental / RENTALL — per-vehicle license

each agency is an isolated island · no shared data · linear, never stronger



RENTY — a network with a hub

each new agency makes the shield & the fleet stronger for all → flywheel



The six new rules (each turns a wound into a moat)

R1 · Trust is a protocol, not a deposit

Risk-scored dynamic / zero deposit. The deposit becomes a priced risk instrument.

R2 · "RENTY Verified" inspection

Dual-signed, hash-chained check-in/out.

Ends damage disputes on both sides.

R3 · Fraud & identity network ★ moat

Fraudster who burns A is flagged for B,C,D.

Impossible for island competitors.

R4 · Embedded insurance + finance

Monetize the #1 cost: waivers, per-rental cover, fleet financing on utilization data.

R5 · Fleet liquidity network

Idle car at A fulfills demand at B in-city.

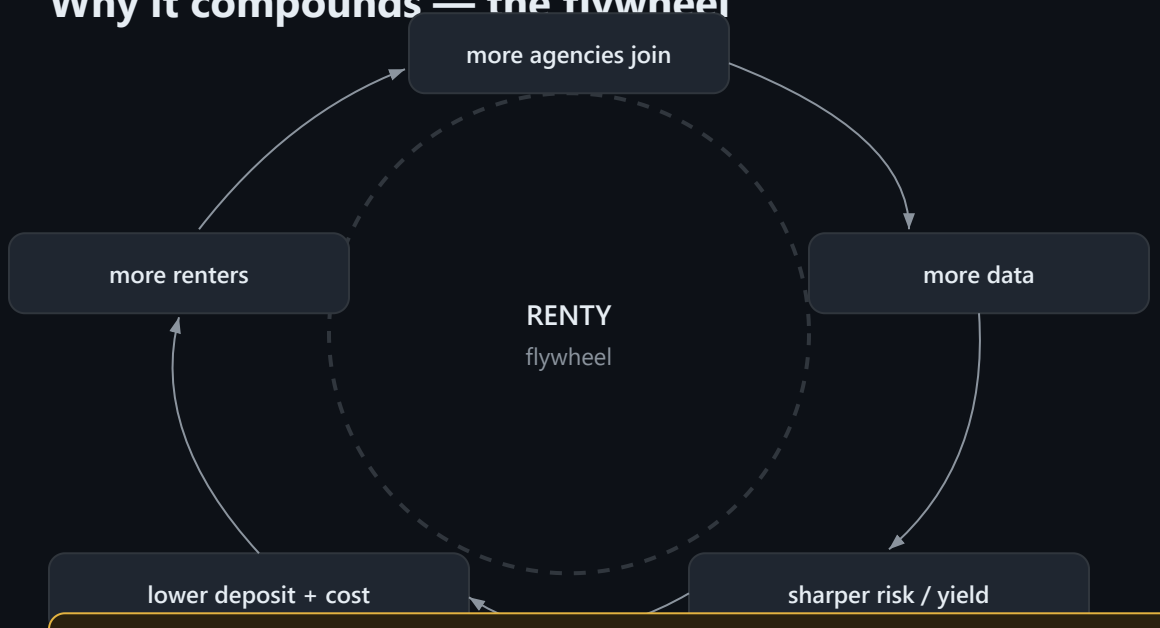
Fragmented fleet → one virtual fleet.

R6 · Own the consumer trust brand

Cash-first. Portable renter reputation.

Relationship survives the signature.

Why it compounds — the flywheel



Sequencing — earn the network

Phase 1 · Win the single agency

Doc 1 + R1/R2 — its own fraud & deposit losses drop. No network yet.

Phase 2 · Switch on the network

R3 fraud graph + R5 liquidity — value that only exists at scale.

Phase 3 · Own the economics

R4 insurance/finance + R6 brand — RENTY becomes the rails.

⚠ R3 deliberately punctures tenant isolation — it stays legal only on explicit consent + lawful basis + crypto-shreddable identity. Moat design = compliance design.